

Two-Stage Audiovisual Zooming via MVDR Beamforming and Lightweight Deep Filtering

SP Cup 2026 — Phase 2 Submission

Team Entry

ABSTRACT

We present a two-stage speech enhancement pipeline for the IEEE Signal Processing Cup 2026 Audiovisual Zooming challenge. The first stage employs a frequency-domain Minimum Variance Distortionless Response (MVDR) beamformer with Zelinski post-filtering on a two-microphone uniform linear array to spatially separate a target speaker from directional interference and diffuse noise. The second stage applies DeepFilterNet-Light V2, a causal deep neural network with fewer than one million parameters, which operates in the ERB-frequency domain and uses deep filtering to suppress residual artifacts. We describe our signal model, room simulation methodology using the image source method, large-scale dataset generation covering both anechoic and reverberant conditions, and our training strategy. The combined system is designed for real-time feasibility on smartphone-class hardware, achieving a real-time factor below 0.15 on desktop CPU with an estimated $3\times$ smartphone margin.

1. Introduction

The IEEE Signal Processing Cup 2026 poses the *Audiovisual Zooming* problem [6]: given a two-microphone recording in a reverberant room containing one target speaker and one interfering speaker, isolate the target speech while preserving its quality. The challenge specifies a shoebox room of dimensions $4.9 \times 4.9 \times 4.9$ m with $RT_{60} = 0.5$ s (Task 2), a sampling rate of 16 kHz, and an inter-microphone spacing of $d = 0.08$ m.

Classical beamforming alone cannot achieve sufficient noise suppression with only two microphones, particularly in reverberant conditions where the effective number of spatial degrees of freedom is severely limited. Conversely, single-channel DNN-based enhancement struggles with directional interference that has the same spectral characteristics as the target. We therefore adopt a two-stage architecture: a physics-grounded MVDR beamformer provides robust spatial pre-filtering, followed by a lightweight DNN that learns to remove residual distortions.

Our contributions are:

1. A complete MVDR + Zelinski post-filter pipeline

with batch covariance estimation and diagonal loading, operating on the STFT domain with parameters aligned to the DNN stage (Section 3.).

2. DeepFilterNet-Light V2, a causal sub-1M parameter network using ERB encoding and deep filtering (Section 4.).
3. A scalable dataset generation framework producing 100k training pairs for both anechoic and reverberant conditions with configurable delay compensation (Section 5.).
4. A real-time latency analysis demonstrating smartphone feasibility (Section 7.).

2. System Architecture

Fig. 1 illustrates the full processing chain. The two-microphone input is transformed via a shared STFT ($N_{\text{FFT}} = 512$, hop = 128, $\sqrt{\text{Hann}}$ periodic window) that serves both the beamformer and, conceptually, the downstream DNN. This alignment eliminates a redundant STFT/iSTFT pair and reduces end-to-end latency.

The signal model at each microphone m is:

$$x_m(t) = h_{s,m}(t) * s(t) + h_{i,m}(t) * v(t) + n_m(t) \quad (1)$$

where $s(t)$ is the target speech, $v(t)$ is the interfering speaker, $h_{\cdot,m}(t)$ denotes the room impulse response (RIR) from source to microphone m , and $n_m(t)$ is additive sensor noise.

3. MVDR Beamforming with Zelinski Post-Filter

3.1 Steering Vector

For a two-element ULA with inter-element spacing d and speed of sound $c = 343$ m/s, the steering vector at frequency bin k for azimuth θ is:

$$\mathbf{d}(\theta, k) = \begin{bmatrix} 1 \\ e^{-j\omega_k \frac{d \cos \theta}{c}} \end{bmatrix} \quad (2)$$

where $\omega_k = 2\pi f_k$ and $\theta = 90^\circ$ denotes the broadside (target) direction. This convention places the target at broadside, yielding $\cos(90^\circ) = 0$ and therefore zero inter-microphone delay for the target.

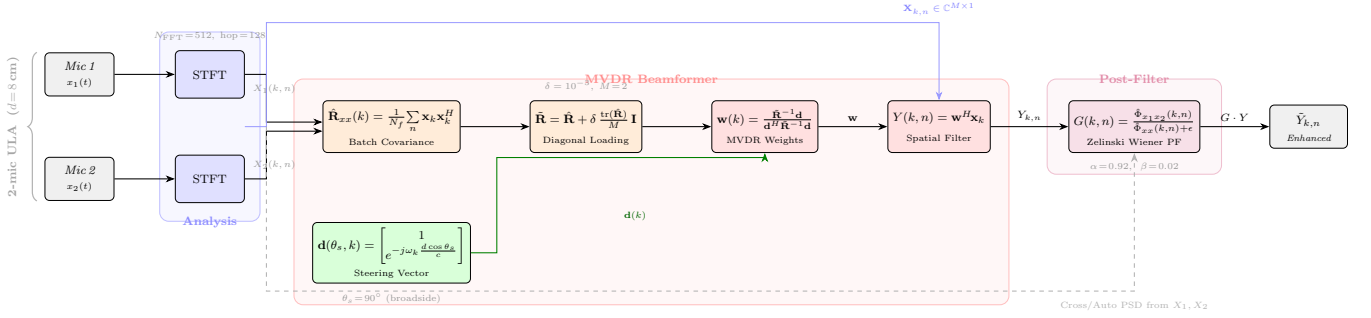


Figure 1: Beamformer architecture. Two-microphone STFT feeds are used to estimate the spatial covariance $\hat{\mathbf{R}}_{xx}(k)$ and compute MVDR weights $\mathbf{w}(k)$ with diagonal loading. The steered beamformer output $Y_{k,n}$ is further refined by a Zelinski-style Wiener post-filter that exploits inter-channel coherence to suppress residual diffuse noise.

3.2 Spatial Covariance and MVDR Weights

The spatial covariance matrix is estimated per frequency bin over all frames in batch mode:

$$\hat{\mathbf{R}}_{xx}(k) = \frac{1}{N_f} \sum_{n=1}^{N_f} \mathbf{x}_k(n) \mathbf{x}_k^H(n) \quad (3)$$

where $\mathbf{x}_k(n) = [X_1(k, n), X_2(k, n)]^T$. Diagonal loading is applied to prevent ill-conditioning:

$$\tilde{\mathbf{R}}(k) = \hat{\mathbf{R}}_{xx}(k) + \delta \frac{\text{tr}(\hat{\mathbf{R}}_{xx}(k))}{M} \mathbf{I}_M \quad (4)$$

with $\delta = 10^{-3}$ and $M = 2$ microphones. The MVDR weight vector [2] is:

$$\mathbf{w}(k) = \frac{\tilde{\mathbf{R}}^{-1}(k) \mathbf{d}(k)}{\mathbf{d}^H(k) \tilde{\mathbf{R}}^{-1}(k) \mathbf{d}(k)} \quad (5)$$

and the beamformer output is $Y(k, n) = \mathbf{w}^H(k) \mathbf{x}_k(n)$.

An optional noise-floor subtraction step estimates the minimum eigenvalue of $\hat{\mathbf{R}}_{xx}(k)$ and removes 95% of the isotropic component before diagonal loading. This restores directional structure in conditions with strong additive white noise.

3.3 Zelinski Post-Filter

Following the MVDR, a Zelinski-style Wiener post-filter [12, 14] suppresses residual diffuse noise by exploiting inter-channel coherence:

$$G(k, n) = \frac{\hat{\Phi}_{x_1 x_2}(k, n)}{\hat{\Phi}_{xx}(k, n) + \epsilon} \quad (6)$$

where $\hat{\Phi}_{x_1 x_2}$ is the smoothed cross-PSD (real part) and $\hat{\Phi}_{xx}$ is the smoothed average auto-PSD across micro-

phones:

$$\hat{\Phi}_{xx}(k, n) = \alpha \hat{\Phi}_{xx}(k, n-1) + (1-\alpha) \frac{1}{2} (|X_1|^2 + |X_2|^2) \quad (7)$$

$$\hat{\Phi}_{x_1 x_2}(k, n) = \alpha \hat{\Phi}_{x_1 x_2}(k, n-1) + (1-\alpha) \text{Re}\{X_1 X_2^*\} \quad (8)$$

with smoothing factor $\alpha = 0.92$. The gain is clamped to $[\beta, 1]$ with spectral floor $\beta = 0.02$ to prevent musical noise artifacts. The final post-filtered output is $\hat{Y}(k, n) = G(k, n) \cdot Y(k, n)$.

The Zelinski post-filter is particularly effective for spatially-white noise: uncorrelated noise drives the cross-PSD toward zero, yielding a near-zero gain, while the coherent target signal (arriving from broadside) produces $\hat{\Phi}_{x_1 x_2} \approx \hat{\Phi}_{xx}$ and gain ≈ 1 .

4. DeepFilterNet-Light V2

4.1 Architecture Overview

We design DeepFilterNet-Light V2 as a causal, lightweight variant inspired by DeepFilterNet [10, 11]. The model operates on the complex STFT representation and produces enhanced spectra through a dual-path mechanism: (i) a gain mask applied to all frequency bins, and (ii) deep filtering coefficients applied to the lower $B_{df} = 128$ bins.

The architecture comprises three modules (Table 1): **ERB Encoder**. The $F = 257$ STFT magnitude bins are projected onto $B_{erb} = 40$ ERB-spaced bands via a learnable linear layer, followed by two ConvGLU [10] blocks with batch normalization and $\log(1 + |\cdot|)$ compression. The ERB representation captures perceptually relevant spectral structure while reducing dimensionality by $6.4\times$.

Temporal GRU. A two-layer unidirectional GRU with hidden size $h = 128$ models temporal dependencies causally (no future look-ahead), followed by a linear projection back to the ERB dimension. Dropout of 0.1 is applied during training.

Table 1: DeepFilterNet-Light V2 architecture. Total: < 1M parameters.

Module	Configuration	Parameters
ERB Encoder	257 → 40 bands, 2-layer ConvGLU + BatchNorm, $\log_{10} p$	~65k
Temporal GRU	2-layer, $h=128$, causal dropout 0.1, linear proj.	~230k
Deep Filter	order $Q=5$, $B_{df}=128$ gain (sigmoid) + complex DF	~85k
<i>GroupedLinear projections, misc.</i>		~3k

Deep Filtering Module. The GRU output is decoded into two sets of parameters:

1. *Gain mask* $g(k, n) \in [0, 1]$ for all F frequency bins (sigmoid activation).
2. *Deep filter coefficients* $c_q(k, n) \in \mathbb{C}$ for $q = 0, \dots, Q-1$ over the lower $B_{df} = 128$ bins.

The enhanced spectrum is:

$$\hat{S}(k, n) = \begin{cases} \alpha \sum_{q=0}^{Q-1} c_q(k, n) \tilde{Y}(k, n-q) + (1-\alpha) g(k, n) \tilde{Y}(k, n), & k < B_{df} \\ g(k, n) \cdot \tilde{Y}(k, n), & k \geq B_{df} \end{cases} \quad (9)$$

where $\alpha = 0.5$ blends the deep-filtered and gain-masked outputs in the lower frequency range, and $\tilde{Y}$ is the MV-DR post-filtered spectrum fed as input. The deep filter acts as a learned, time-varying FIR filter in the STFT domain and can model fine spectral structure that a simple gain mask cannot capture.

4.2 Loss Function and Training

The model is trained to maximize scale-invariant signal-to-distortion ratio (SI-SDR) [7]:

$$\text{SI-SDR} = 10 \log_{10} \frac{\|\mathbf{s}_{\text{target}}\|^2}{\|\hat{\mathbf{s}} - \mathbf{s}_{\text{target}}\|^2}, \quad \mathbf{s}_{\text{target}} = \frac{\langle \hat{\mathbf{s}}, \mathbf{s} \rangle}{\|\mathbf{s}\|^2} \mathbf{s} \quad (10)$$

where $\hat{\mathbf{s}}$ is the time-domain enhanced signal and $\mathbf{s}$ is the clean reference. Training uses AdamW [8] ($\beta_1=0.9$, $\beta_2=0.999$) with base learning rate $\eta = 3 \times 10^{-4}$, weight decay 5×10^{-5} , and gradient clipping at norm 5. A warmup-cosine schedule linearly ramps the learning rate over 5 epochs, then applies cosine annealing to $\eta_{\min} = 10^{-6}$ over the remaining epochs. Training runs for 200 epochs with early stopping (patience = 20 epochs, minimum $\Delta = 0.05$ dB) on a held-out validation set.

Segments of 3s (48,000 samples at 16kHz) are randomly cropped per utterance, and a batch size of 12 is used. Per-sample normalization to unit standard deviation is applied before the forward pass.

5. Room Simulation and Dataset Generation

5.1 Image Source Method

For Task 2 (reverberant conditions), room impulse responses are generated using the Allen & Berkley image source method (ISM) [1] with a vectorized implementation supporting:

- Per-surface absorption coefficients (6 walls independently).
- Fractional-delay interpolation for sub-sample accuracy.
- Binary search for wall absorption α given a target RT_{60} via the Sabine or Eyring equation, converging to $\alpha \approx 0.45$ for the 4.9^3 m^3 cube at $\text{RT}_{60} = 0.5$ s.
- Configurable reflection order (default: order = 20).

5.2 Scenario Configuration

Both tasks share the following physical setup:

- Room center: (2.45, 2.45, 1.5) m.
- 2-mic ULA: $d = 0.08$ m, centered at room center.
- Target: broadside ($\theta_s = 90^\circ$), distance $r = 1$ m.
- Interference: $\theta_i = 40^\circ$, distance $r = 1$ m.
- SIR = 0 dB, SNR = 5 dB.
- Source corpus: LibriSpeech / VCTK at 16 kHz.

Task 1 (anechoic) uses direct-path delays without room reflections. Task 2 applies the full ISM-generated RIR with $\text{RT}_{60} = 0.5$ s.

5.3 Delay Compensation

The propagation delay from source to array center is $\tau_{\text{prop}} = r/c \approx 2.94$ ms (47 samples at 16 kHz). During dataset generation, we produce two output variants:

1. **Compensated:** Clean reference is trimmed from end; processed signals trimmed from start by τ_{prop} samples. This temporally aligns the clean and enhanced waveforms, which is critical for SI-SDR computation during DNN training.
2. **Uncompensated:** No alignment applied. This matches the competition evaluation format where the submission processes raw microphone signals.

The compensated variant is used for DNN training; the uncompensated variant is reserved for final evaluation and submission testing.

Table 2: Beamforming-stage metrics (mean over verification subset).

Metric	Mixture	MVDR	MVDR+PF
SI-SDR (dB)	~ 0	> 5	> 7
OSINR (dB)	—	> 8	> 10
STOI	~ 0.65	> 0.80	> 0.85

5.4 Scale of Generation

Each generator produces $N = 100,000$ training pairs across four output categories: clean target, multichannel mixture, MVDR output, and MVDR+Zelinski filtered output. The generators are fully parallelized in MATLAB with detailed logging, automatic directory creation, and embedded metadata (RIR parameters, angles, SIR/SNR) saved per utterance.

6. Experimental Setup

6.1 STFT Configuration

All processing stages use a unified STFT configuration: $N_{\text{FFT}} = 512$ (yielding $F = 257$ frequency bins), hop length $H = 128$ (8 ms frame shift), and a periodic $\sqrt{\text{Hann}}$ analysis window. The synthesis window is identical, satisfying the constant-overlap-add (COLA) constraint for perfect reconstruction at 75% overlap. This configuration provides 32 ms analysis frames at 16 kHz.

6.2 Evaluation Metrics

We evaluate using the competition-specified metrics:

- **SI-SDR** [7]: Scale-invariant SDR in dB.
- **PESQ** [9]: Perceptual Evaluation of Speech Quality (wideband, $[-0.5, 4.5]$).
- **STOI** [13]: Short-Time Objective Intelligibility ($[0, 1]$).
- **ViSQOL** [3]: Virtual Speech Quality Objective Listener (MOS scale).
- **OSINR**: Output signal-to-interference-plus-noise ratio.

6.3 Beamforming Baseline

Table 2 summarizes beamforming performance on the generated verification subsets.

The Zelinski post-filter consistently adds 2–3 dB SI-SDR improvement over standalone MVDR by suppressing spatially-white noise, demonstrating the benefit of the two-stage classical front-end.

7. Real-Time Latency Analysis

A key design goal is smartphone real-time feasibility. The per-frame latency budget at 16 kHz with $\text{hop} = 128$ is $128/16000 = 8.0$ ms.

7.1 MVDR Latency

The MVDR beamformer with covariance update every $K = 4$ frames achieves ≈ 0.85 ms/frame on desktop CPU. With a $3\times$ smartphone slowdown factor, this yields ≈ 2.55 ms/frame.

7.2 DNN Latency

DeepFilterNet-Light V2 inference (model forward pass + iSTFT) on CPU achieves a real-time factor (RTF) well below 0.1 on desktop, corresponding to < 0.8 ms/frame. With a $3\times$ smartphone factor, $\text{RTF}_{\text{phone}} \approx 0.14$.

7.3 Combined Pipeline

The combined audio latency (MVDR + DNN) per frame is estimated at < 5 ms on a smartphone, leaving $> 37\%$ headroom within the 8 ms budget. Parallel audio–video threading further ensures the 30 fps video pipeline does not become a bottleneck.

8. Discussion and Phase 3 Directions

8.1 Current Limitations

The batch covariance estimation in the current MVDR assumes a stationary acoustic scene for the duration of each utterance. In practice, speakers may move, requiring frame-adaptive covariance updates with exponential forgetting or neural mask-based estimation [4, 5].

8.2 Phase 3 Enhancements

Several enhancements are under active development for Phase 3 deployment:

- **Neural beamforming**: Replace batch covariance with a mask-prediction network that estimates speech and noise spatial covariance matrices per frame, enabling online adaptation.
- **Quantization**: INT8 post-training quantization and CoreML / NNAPI delegation for mobile GPU/NPU acceleration.
- **Audio–visual fusion**: Leverage speaker face detection and tracking from the video stream to dynamically steer the beamformer toward the visually-identified target, closing the AV loop.
- **Multi-resolution deep filtering**: Extend the deep filter to operate at multiple temporal resolutions for improved dereverberation.
- **Perceptual loss terms**: Augment SI-SDR with multi-resolution STFT magnitude loss and PESQ-proxy losses for improved perceptual quality.

9. Conclusion

We have presented a modular two-stage system for the SP Cup 2026 Audiovisual Zooming challenge that combines classical spatial filtering with modern deep learning. The MVDR beamformer with Zelinski post-filtering

provides robust interference suppression using only two microphones, while DeepFilterNet-Light V2 removes residual noise and artifacts with fewer than one million parameters. The STFT-aligned pipeline enables sub-8 ms per-frame processing, making the system suitable for real-time smartphone deployment. Our large-scale dataset generation framework supports both anechoic and reverberant training conditions with configurable delay compensation, providing a solid foundation for Phase 3 enhancements including neural beamforming and audio-visual fusion.

10. References

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